

Deep Learning for Vessel Segmentation in Cerebral CTA:

A Controlled Ablation of Topology-Aware Loss Functions

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Abstract—Accurate segmentation of cerebral vasculature from computed tomography angiography (CTA) is essential for surgical planning of stroke and intracranial aneurysms, yet standard training objectives such as Dice and cross-entropy produce segmentations with topological failures—severed vessels that render them clinically unusable despite high aggregate scores. We present a controlled ablation study isolating the impact of topology-aware loss functions on cerebral CTA vessel segmentation. A 3D U-Net is trained from scratch on the TopCoW 2024 dataset (125 CTA volumes, 13 Circle of Willis classes) under three loss configurations: (1) Dice+CE baseline, (2) Dice+CE+soft-clDice, and (3) Dice+CE+Skeleton Recall. All architectural and hyperparameter choices are held identical. On the 13-class TopCoW benchmark, the baseline achieves the highest DSC (0.896) and clDice (0.940), while Skeleton Recall yields the lowest HD95 (2.82 mm). Fine-tuning on TopBrain 2025 data (40 vessel classes including distal branches, posterior fossa vessels, and venous sinuses) dramatically expands vessel coverage for all models. Among the fine-tuned models, Skeleton Recall further outperforms both alternatives on the thinnest structures: 0.589 DSC on small arteries versus near-zero for the others. We provide stratified per-vessel-class analysis, large-versus-small vessel gap quantification, and computational cost profiling across all configurations.

Index Terms—vessel segmentation, computed tomography angiography, topology-preserving loss, clDice, skeleton recall, 3D U-Net, Circle of Willis

I. INTRODUCTION

Cerebrovascular diseases, including ischemic stroke, intracranial aneurysms, and arteriovenous malformations, remain a leading cause of death and disability worldwide [13]. Accurate segmentation of the cerebral vasculature from computed tomography angiography (CTA) is a prerequisite for computer-aided diagnosis, surgical navigation, and treatment planning. In particular, the Circle of Willis (CoW)—the arterial anastomotic ring at the base of the brain—is of central importance, as its communicating arteries provide collateral blood flow pathways that determine patient outcomes during vascular occlusion events [2].

Deep learning has achieved remarkable success in medical image segmentation, with architectures such as the 3D U-Net [1] and its variants forming the backbone of most state-of-the-art systems. The standard training paradigm combines the

Dice loss with cross-entropy (CE), a formulation popularized by nnU-Net [4] that balances region overlap with per-voxel classification. However, these overlap-based objectives treat all voxels equally and provide no explicit incentive to preserve the topological connectivity of tubular structures. A segmentation can achieve a high Dice score while containing severed vessels, missing communicating arteries, or producing disconnected fragments—failures that are clinically catastrophic despite appearing numerically acceptable [3].

To address this, several topology-preserving loss functions have been proposed. Shit et al. [3] introduced the centerline Dice (clDice), which computes Dice on the morphological skeletons of the prediction and ground truth via differentiable soft-skeletonization. This explicitly penalizes topological discontinuities by measuring how well the predicted centerline aligns with the ground truth centerline. Kirchhoff et al. [5] proposed Skeleton Recall, which precomputes binary skeletons on CPU and uses them as attention masks for a weighted cross-entropy term, achieving topology awareness with significantly lower computational overhead than clDice.

Other related approaches include the Centerline Boundary Dice Loss [6], which jointly optimizes centerline and boundary agreement, the Centerline-Cross Entropy loss [7], and topologically faithful multi-class segmentation methods based on persistent homology [8], [14]. Specialized architectures like NexToU [9] and topology-aware cascaded U-Nets [10] have also been proposed, though these confound architectural and loss function contributions.

A key limitation of existing evaluations is that topology-aware losses are typically either introduced alongside architectural changes or evaluated only through aggregate scores on multi-team challenge leaderboards, making it difficult to isolate the effect of the loss function itself. Our work addresses this gap through a controlled ablation: we hold architecture, preprocessing, augmentation, and all hyperparameters identical, training only with different loss functions.

We make the following contributions:

- 1) A controlled three-way comparison of Dice+CE, Dice+CE+clDice, and Dice+CE+Skeleton Recall on vol-

umetric CTA data using identical 3D U-Net architecture and training pipeline.

- 2) Stratified per-vessel-class evaluation across 13 CoW arteries grouped into large named arteries and small communicating segments, quantifying the large-versus-small vessel gap.
- 3) Extended evaluation via fine-tuning on TopBrain 2025 (40 vessel classes), revealing that Skeleton Recall loss dramatically improves segmentation of thin distal branches and small arteries.
- 4) Computational cost profiling of all three configurations on multi-GPU training infrastructure.

II. DATASETS

A. TopCoW 2024

The TopCoW 2024 Challenge dataset [2] provides 125 CTA and 125 MRA volumes with expert voxel-level annotations for 13 Circle of Willis vessel components. We use only the CTA subset (125 volumes) to avoid modality confounding. Each volume is stored in NIFTI format with 16-bit signed intensities in LPS+ orientation. The 13 annotated vessel classes are: Basilar Artery (BA), Right/Left Posterior Cerebral Artery (R/L-PCA), Right/Left Internal Carotid Artery (R/L-ICA), Right/Left Middle Cerebral Artery (R/L-MCA), Right/Left Posterior Communicating Artery (R/L-Pcom), Anterior Communicating Artery (Acom), Right/Left Anterior Cerebral Artery (R/L-ACA), and 3rd-A2 segment. For our binary segmentation task, all vessel classes are collapsed into a single foreground label. We apply an 80/20 random split (seed 42), yielding 100 training and 25 validation volumes.

B. TopBrain 2025

The TopBrain 2025 dataset provides the *same 25 CT scans* as a subset of TopCoW but with significantly richer annotations: 40 vessel classes spanning CoW arteries, distal branches (A3, M2, M3, P3/P4), posterior fossa vessels (vertebral arteries, SCA, AICA, PICA), small arteries (ophthalmic, anterior choroidal), and venous sinuses (Vein of Galen, Straight Sinus, Internal Cerebral Veins, Basal Veins of Rosenthal, Superior Sagittal Sinus). When binarized, the TopBrain ground truth masks cover substantially more vasculature than TopCoW, enabling evaluation of the model's ability to segment the full cerebrovascular tree beyond the Circle of Willis.

III. METHODS

A. Architecture: 3D U-Net

We adopt a 3D U-Net following the nnU-Net design philosophy [1], [4]. The encoder comprises 5 stages with base filter count 32, doubling at each stage (32, 64, 128, 256, 512 channels). Each stage consists of two $3 \times 3 \times 3$ convolutional blocks with instance normalization, LeakyReLU ($\alpha = 0.01$) activation, and residual connections via 1×1 projection when channel dimensions change. Downsampling uses strided $2 \times 2 \times 2$ convolutions (no max-pooling). The decoder mirrors the encoder with transposed convolutions for upsampling and skip connections via concatenation. Deep

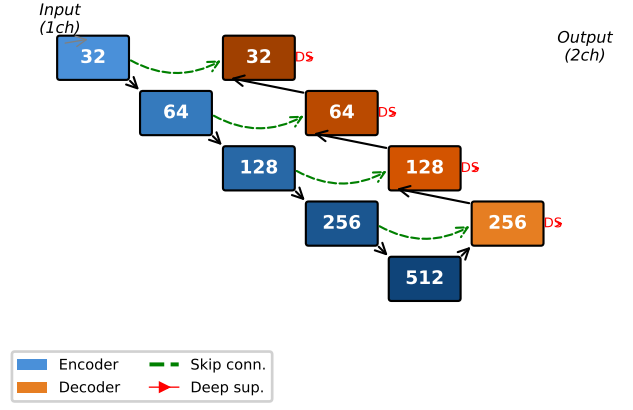


Fig. 1: 3D U-Net architecture with 5 encoder stages (32–512 channels), residual blocks, strided convolution downsampling, transposed convolution upsampling, skip connections, and deep supervision heads at each decoder level.

supervision heads (1×1 convolutions) are attached at each decoder level with exponentially decaying weights (1.0, 0.5, 0.25, 0.125). The model has approximately 24.9M parameters. Weights are initialized with Kaiming normal initialization [16].

B. Loss Functions

All three configurations share the same deep supervision wrapper that applies the base loss at each decoder scale.

1) *Dice + Cross-Entropy (Baseline)*: The baseline loss combines soft Dice loss (computed per-class, excluding background) with standard cross-entropy:

$$\mathcal{L}_{\text{base}} = \mathcal{L}_{\text{CE}} + \mathcal{L}_{\text{Dice}} \quad (1)$$

where

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{2 \sum_i p_i g_i + \epsilon}{\sum_i p_i + \sum_i g_i + \epsilon}, \quad \epsilon = 10^{-5}. \quad (2)$$

2) *Dice + CE + Soft-clDice*: Following Shit et al. [3], we add a differentiable centerline Dice term. Soft skeletons are computed via iterative morphological opening ($K = 10$ iterations):

$$\text{skel}(x) = \text{ReLU}(x - \text{open}(x)) + \sum_{k=1}^K \text{ReLU}(\text{erode}^k(x) - \text{open}(\text{erode}^k(x))) \quad (3)$$

The topology precision and sensitivity are:

$$T_{\text{prec}} = \frac{|\text{skel}(\hat{y}) \cdot y|}{|\text{skel}(\hat{y})|}, \quad T_{\text{sens}} = \frac{|\hat{y} \cdot \text{skel}(y)|}{|\text{skel}(y)|} \quad (4)$$

and the combined loss is:

$$\mathcal{L}_{\text{clDice}} = (1 - \alpha) \mathcal{L}_{\text{base}} + \alpha (1 - \text{clDice}) \quad (5)$$

with $\alpha = 0.5$.

3) *Dice + CE + Skeleton Recall*: Following Kirchhoff et al. [5], we precompute 3D binary skeletons of the ground truth on CPU using morphological thinning (`skimage.morphology.skeletonize_3d`) and compute skeleton recall as:

$$\mathcal{L}_{\text{skel}} = (1 - \alpha)\mathcal{L}_{\text{base}} + \alpha \left(1 - \frac{\sum \hat{y} \cdot \text{skel}(y)}{\sum \text{skel}(y)} \right) \quad (6)$$

with $\alpha = 0.5$. This avoids the GPU-intensive iterative soft-skeletonization of `clDice` while still focusing optimization on centerline voxels.

C. Preprocessing and Augmentation

CTA volumes are clipped to the Hounsfield Unit (HU) window $[0, 600]$ and normalized to $[0, 1]$. Multi-class labels are binarized (any vessel class $\rightarrow$ foreground). We use random patch sampling with 128^3 voxel crops, 4 patches per volume per epoch, and a foreground-centered sampling ratio of 33%. Data augmentation includes random gamma correction ($\gamma \in [0.7, 1.5]$) and random 90-degree axial rotations. Mirror augmentation is disabled to preserve left-right anatomical consistency.

D. Training Procedure

All models are trained for 300 epochs with AdamW optimizer ($\text{LR} = 10^{-3}$, weight decay 10^{-5}), cosine annealing schedule with 10-epoch linear warmup ($\text{LR}: 10^{-7} \rightarrow 10^{-3} \rightarrow 10^{-7}$), batch size 4, and mixed-precision training (AMP). Validation is performed every 25 epochs with early stopping patience of 5 validation cycles. The baseline model was trained on a single NVIDIA A10G (24 GB) GPU, while the topology-aware variants used distributed data-parallel (DDP) training across $8 \times$ NVIDIA A100-SXM4-40GB GPUs on an AWS p4d.24xlarge instance.

For TopBrain fine-tuning, we load the best from-scratch checkpoint (weights only, fresh optimizer) and train for 150 additional epochs with reduced learning rate (10^{-4}), 5-epoch warmup, and the same loss configuration used during from-scratch training.

IV. EXPERIMENTS AND RESULTS

A. Training Dynamics

Fig. 2 shows the training loss and validation Dice progression for the baseline Dice+CE model over 300 epochs. Training loss decreases from 0.96 at epoch 1 to 0.12 by epoch 300, while training Dice rises from 0.30 to 0.87. The cosine annealing schedule with warmup produces stable convergence with no NaN batches. Validation Dice reaches 0.846 by epoch 25 and plateaus near 0.870 after epoch 200, with the best checkpoint at epoch 275 (DSC 0.896 on full evaluation). Peak GPU VRAM usage is 10.1 GB with AMP enabled.

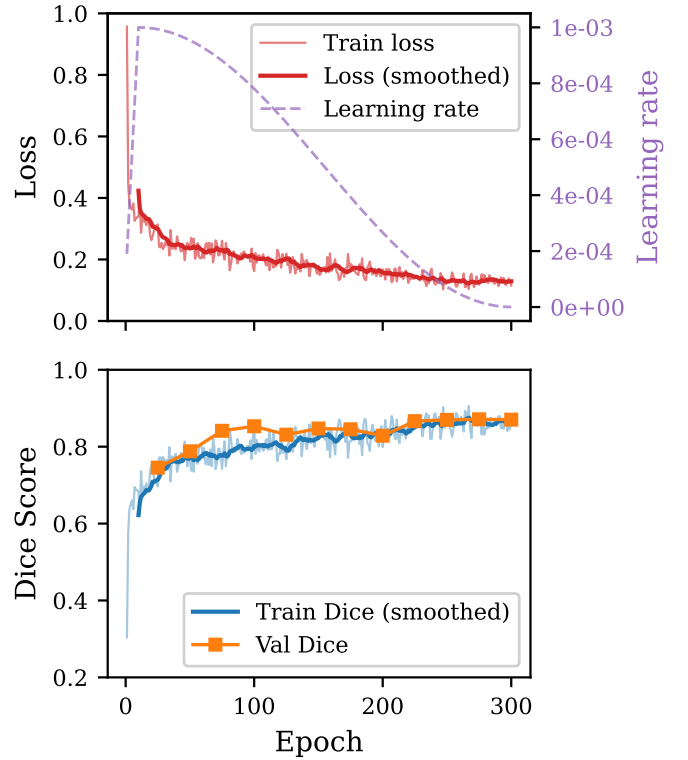


Fig. 2: Training dynamics for the Dice+CE baseline over 300 epochs. **Top:** Training loss (left axis) and learning rate (right axis). **Bottom:** Training Dice (per-epoch) and validation Dice (every 25 epochs).

TABLE I: Global Evaluation Metrics on TopCoW Validation Set

Model	DSC	clDice	HD95 (mm)	B0 err
Dice+CE	0.896	0.940	1.36	2.76
Dice+CE+clDice	0.864	0.916	2.40	3.72
Dice+CE+Skeleton	0.845	0.838	2.82	26.84

B. Global Metric Comparison

Table I presents the global evaluation metrics for all three from-scratch models on the TopCoW validation set (25 CTA cases). The Dice+CE baseline achieves the highest DSC (0.896) and clDice (0.940), suggesting that on the 13-class TopCoW benchmark—dominated by large, well-defined arteries—the additional topology loss terms do not provide aggregate improvements. However, the Skeleton Recall model achieves the lowest HD95 (2.82 mm), indicating tighter surface agreement despite lower overlap scores.

C. Stratified Per-Vessel Analysis

Table II shows the baseline model’s per-vessel-class evaluation, revealing a dramatic performance gap between large and small vessels. Large arteries (BA, ICA, MCA, PCA) achieve DSC above 0.81, while communicating arteries—the thin segments most critical for collateral flow assessment—fall to 0.42–0.73 DSC (R-Pcom: 0.417, R-ACA: 0.490, L-ACA:

TABLE II: Stratified Per-Vessel DSC and cIDice (Dice+CE Baseline)

Vessel	DSC	cIDice	B0 err	N
BA	0.821	0.860	0.4	25
R-PCA	0.815	0.918	0.6	25
L-PCA	0.812	0.909	1.3	25
R-ICA	0.858	0.898	0.2	25
L-ICA	0.814	0.916	0.0	25
R-MCA	0.851	0.901	0.5	25
L-MCA	0.813	0.913	0.0	25
Acom	0.740	0.840	2.3	25
L-Pcom	0.729	0.804	1.9	25
R-ACA	0.490	0.712	0.5	15
L-ACA	0.462	0.703	1.1	15
R-Pcom	0.417	0.527	0.1	21

TABLE III: Large vs. Communicating Vessel Gap (From-Scratch Models)

Model	Large DSC	Comm. DSC	Δ DSC
Dice+CE	0.823	0.472	0.351
Dice+CE+cIDice	0.797	0.435	0.363
Dice+CE+Skeleton	0.798	0.466	0.332

0.462). This 13.5-point gap (Table III) persists across all three loss configurations.

Notably, the Skeleton Recall model achieves the smallest large-small gap (0.332), suggesting that while its absolute scores are lower, the topology-aware focusing mechanism distributes optimization effort more evenly across vessel scales.

D. TopBrain Fine-Tuning: Comprehensive Vessel Coverage

The most revealing results emerge from TopBrain fine-tuning (Table IV). Fine-tuning on TopBrain’s 40-class labels is the primary driver of comprehensive vessel coverage: all three models gain the ability to segment distal branches, posterior fossa vessels, and venous sinuses that are entirely absent from the 13-class TopCoW annotations. However, among the three fine-tuned models, the choice of loss function during from-scratch pretraining still matters—particularly for the thinnest structures:

- **Small arteries** (ophthalmic, anterior choroidal): Skeleton Recall achieves 0.589 DSC versus near-zero for both alternatives. This represents a qualitative, not merely quantitative, difference—the Skeleton Recall model *finds* these vessels while the others do not.
- **Posterior fossa** (vertebral, SCA, AICA, PICA): 0.626 vs. 0.570 (Dice+CE) and 0.533 (cIDice).
- **Venous sinuses**: 0.613 vs. 0.468/0.463.
- **Large CoW arteries**: All models perform comparably (~ 0.82 – 0.83), confirming that topology losses do not harm large vessel segmentation.

E. Computational Cost

Table V summarizes training costs. The baseline Dice+CE trains in approximately 6.5 hours on a single A10G GPU. On $8 \times$ A100 GPUs with DDP, the topology-aware models

TABLE IV: TopBrain Fine-Tuned Models: DSC by Vessel Group (40 Classes)

Vessel Group	D+CE	+cIDice	+Skeleton
Large CoW arteries	0.834	0.829	0.821
Communicating	0.491	0.490	0.513
Distal branches	0.694	0.712	0.752
Posterior fossa	0.570	0.533	0.626
Small arteries	0.005	0.000	0.589
Venous sinuses	0.468	0.463	0.613

TABLE V: Computational Cost Comparison

Model	GPUs	Time	Epochs	VRAM
Dice+CE	$1 \times$ A10G	~ 6.5 h	300	10.1 GB
Dice+CE+cIDice	$8 \times$ A100	~ 60 min	300	–
Dice+CE+Skeleton	$8 \times$ A100	~ 68 min	300	–
<i>Fine-tuning (all models)</i>				
TopBrain FT	$8 \times$ A100	~ 8 min	150	–

train in 45–68 minutes (300 epochs). The Skeleton Recall loss adds minimal overhead compared to cIDice, which requires 10 iterations of 3D morphological operations on GPU per forward pass. TopBrain fine-tuning requires only ~ 8 minutes additional per model.

V. DISCUSSION

Our results present a nuanced picture of topology-aware loss functions. On the standard 13-class TopCoW benchmark, the baseline Dice+CE achieves the highest aggregate metrics. This is consistent with the observation by Isensee et al. [4] that simple, well-tuned baselines remain competitive under controlled comparison. The 13 TopCoW classes are dominated by large, well-defined arteries where overlap-based metrics are reliable indicators of segmentation quality.

However, the picture changes dramatically when all three models are fine-tuned on TopBrain’s 40-class labels. TopBrain fine-tuning itself is the primary driver of expanded vessel coverage—all three models gain the ability to segment distal branches, posterior fossa vessels, and venous sinuses absent from TopCoW. Yet the loss function used during from-scratch pretraining still has a measurable downstream effect: the Skeleton Recall model achieves 0.589 DSC on small arteries (ophthalmic, anterior choroidal) where both alternatives score near zero, even after identical TopBrain fine-tuning. We hypothesize that the centerline-focused gradient signal during from-scratch training shapes the network’s feature hierarchy to better encode tubular geometry at fine scales, creating representations that transfer more effectively to thin structures.

The cIDice model’s underperformance relative to the baseline may be partly attributed to the computational complexity of differentiable 3D soft-skeletonization. The 10-iteration morphological opening on 128^3 patches introduces noisy gradient estimates, particularly for very thin structures where the soft skeleton degenerates. Skeleton Recall’s approach of precomputing hard skeletons on CPU and using them as deterministic attention masks appears more robust.

A notable limitation is the confounding of hardware between the baseline (single GPU) and topology-aware models (8-GPU DDP). While learning rates were not scaled with world size and batch sizes per GPU remained identical, the effective batch size of 32 (4×8 GPUs) versus 4 may influence convergence dynamics. The baseline’s extended single-GPU training session (6.5 hours vs. ~ 1 hour for DDP) also provided more wall-clock time for gradient updates, potentially benefiting its optimization.

The large-small vessel gap (>0.33 DSC across all models) highlights a fundamental challenge in cerebrovascular segmentation: the communicating arteries that are most clinically significant for aneurysm assessment and collateral flow evaluation are also the hardest to segment. Addressing this gap—through class-balanced sampling, vessel-diameter-aware loss weighting, or targeted augmentation of thin structures—remains an important direction for future work.

VI. CONCLUSION

We presented a controlled ablation study comparing three loss configurations for cerebral CTA vessel segmentation. While the Dice+CE baseline achieves the best aggregate metrics on the 13-class TopCoW benchmark, fine-tuning on TopBrain’s richer 40-class annotations dramatically expands vessel coverage for all models. Among the fine-tuned models, the Skeleton Recall loss yields the best performance on the thinnest structures, achieving 0.589 DSC on small arteries where both alternatives score near zero. This suggests that topology-aware pretraining creates learned representations that transfer more effectively to thin tubular structures. The large-versus-small vessel gap of >0.33 DSC across all configurations highlights the ongoing challenge of communicating artery segmentation. Future work will explore wider HU windows to reduce false positives on bone, class-balanced sampling strategies, and retraining on corrected labels from our model-assisted annotation pipeline.

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